

REVIEW 01 Automatic Industrial Garbage Collection and Segregation Robot

OVERVIEW

Goon et al. (2021) developed an Automatic Industrial Garbage Collection and Segregation Robot (AIGCSR) aiming to reduce human involvement in industrial waste handling. The system combines a mobile platform, robotic arm, camera monitoring, sensors, Arduino control, and IoT connectivity to collect waste from hard-to-reach locations and separate it into metallic and non-metallic categories. It emphasizes a low-cost solution for resource-constrained contexts like Bangladesh.

PROJECT OBJECTIVES

1. Automate industrial garbage collection using a robotic arm.
2. Monitor waste using a camera-based system and separate into metallic/non-metallic categories.
3. Incorporate IoT-based control for operation while reducing time, labor, and cost.
4. Provide an affordable solution adaptable for local industries.

RESEARCH GAP

Existing systems relied on advanced, expensive technologies impractical for local implementation. Additionally, the proposed prototype is limited strictly to metallic vs. non-metallic sorting. Thus, a gap exists for an integrated, low-cost robotic system capable of physical collection and basic sorting in resource-constrained environments.

RESEARCH QUESTIONS

- **RQ1:** How can a low-cost robotic system automate waste collection with minimal human intervention?
- **RQ2:** How can sensors and mechanisms be integrated to distinguish metallic vs. non-metallic waste?
- **RQ3:** Can IoT control be incorporated without substantially increasing system complexity and cost?
- **RQ4:** How suitable is this approach for resource-constrained environments like Bangladesh?

KEY CONTRIBUTION

Combines physical collection and segregation into a single operational platform using Arduino and IoT, emphasizing affordability over complex AI algorithms.

OVERVIEW

Lakhout (2025) examines smart technologies across municipal waste stages (collection, sorting, recycling, monitoring). It discusses IoT smart bins, GPS tracking, AI route optimization, predictive generation modeling, automated sorting, and blockchain traceability, emphasizing modular adoption for constrained cities.

PROJECT OBJECTIVES

- Improve collection via real-time monitoring and AI route optimization.
- Automate sorting, predict generation patterns, and enhance recycling efficiency.
- Explore blockchain for waste traceability and identify barriers to large-scale adoption.

RESEARCH GAP

Most smart waste frameworks target well-funded urban environments. Technical reliability issues (sensor failure, material misidentification) and the lack of integrated, modular, and affordable AI/IoT architectures present a major gap for resource-limited cities.

RESEARCH QUESTIONS

- **RQ1:** How can AI improve waste collection, sorting, and recycling in urban environments?
- **RQ2:** How can IoT sensors and real-time monitoring improve waste collection efficiency?
- **RQ3:** How can AI/IoT be integrated into scalable architectures for resource-constrained cities?
- **RQ4:** What technical and operational limitations affect system reliability?

KEY CONTRIBUTION

Provides a comprehensive, system-level architecture linking collection, route optimization, sorting, recycling, and blockchain traceability tailored for scalable urban adoption.

OVERVIEW

Fotovvatikhah et al. (2025) conduct a systematic literature review following PRISMA guidelines, analyzing 97 studies (2020–2025). The study categorizes ML, DL, and hybrid models while assessing public datasets, reported accuracy, and real-world deployment challenges.

PROJECT OBJECTIVES

1. Identify public datasets and analyze ML, DL, and hybrid waste classification techniques.
2. Compare classification performance and evaluate dataset quality, standardization, and balance.
3. Outline deployment limitations and future research directions.

RESEARCH GAP

Prior reviews lacked combined coverage of ML, DL, hybrid models, and datasets. A major practical gap exists in training datasets: most are small, imbalanced, or collected under controlled lab conditions, causing poor generalization to real-world waste environments with occlusion and mixed materials.

RESEARCH QUESTIONS

- **RQ1:** What are the publicly available datasets for waste classification?
- **RQ2:** How does AI contribute to waste classification, and what key techniques enhance efficiency?

KEY CONTRIBUTION

Delivers a structured research map identifying dataset limitations, performance trends, and unresolved challenges, proposing lightweight models and improved dataset standardization.

OVERVIEW

González et al. (2025) propose a smart sorting bin integrating computer vision (YOLO), sensors, multi-agent software architecture, and blockchain. Autonomous agents handle detection, sorting decisions, and mechanical actions while maintaining tamper-proof sorting logs on a blockchain.

PROJECT OBJECTIVES

- Develop an affordable, multi-agent smart bin combining computer vision and physical sensors.
- Evaluate YOLO variants for low-cost hardware deployment.
- Implement blockchain for transparent events tracking and operational traceability.

RESEARCH GAP

Bridges the gap between high accuracy models and low-cost, modular hardware deployment. It resolves limitations in system modularity, decision traceability, and high computational demands present in prior architectures.

RESEARCH QUESTIONS

- **RQ1:** How can computer vision and sensors combine to improve automated sorting?
- **RQ2:** How can multi-agent architecture support modular smart-bin operations?
- **RQ3:** How effectively does YOLO operate on constrained hardware?
- **RQ4:** How does blockchain enhance auditability and traceability in waste management?

KEY CONTRIBUTION

Combines multi-agent modular control with blockchain logging and vision models, proving decentralized software design can run effectively on low-cost hardware.

OVERVIEW

Lahoti et al. (2024) present a multi-class waste sorting prototype using YOLOv5 and a 3D-printed robotic arm. The system detects dry waste categories (biodegradable, cardboard, glass, metal, paper, plastic) and controls servo motors via an Arduino to perform physical pick-and-place operations, achieving an 80% success rate in testing.

PROJECT OBJECTIVES

1. Automate multi-class dry-waste segregation using YOLOv5 object detection.
2. Interface AI detection outputs with Arduino servo controllers.
3. Operate a physical gripper arm to sort items into distinct waste bins.

RESEARCH GAP

Addresses the gap between theoretical software classification and physical actuation. Limitations remain regarding background confusion (e.g., paper misclassified as background), GPU training constraints, and performance degradation in non-controlled environments.

RESEARCH QUESTIONS

- **RQ1:** Can YOLOv5 accurately identify multiple dry waste categories for physical manipulation?
- **RQ2:** How can vision predictions be reliably converted to robotic arm motor signals?
- **RQ3:** How effectively can low-cost arms execute pick-and-place sorting?
- **RQ4:** What failure modes occur with visually ambiguous or background-blended items?

KEY CONTRIBUTION

Demonstrates an end-to-end physical prototype linking YOLOv5 detection directly to a 3D-printed robotic arm for physical multi-class sorting.

OVERVIEW

Ahmad et al. (2025) propose a CNN deep learning framework classifying waste into 12 categories. Trained on 15,535 augmented images to address class imbalance, their ResNet-based model achieved a 98.16% classification accuracy.

PROJECT OBJECTIVES

- Classify waste across 12 distinct categories using deep learning architectures.
- Apply data augmentation and balancing strategies to maximize model generalization.
- Evaluate model metrics (precision, recall, F1-score) for real-world smart city deployment.

RESEARCH GAP

Existing deep learning models suffer from class imbalance, inability to handle visually similar or occluded items, and high computational costs that prevent real-time deployment on edge devices.

RESEARCH QUESTIONS

- **RQ1:** How effectively can a CNN classify 12 categories of urban waste from images?
- **RQ2:** To what extent does data augmentation mitigate class imbalance and improve generalization?
- **RQ3:** How robust is the model against visual similarity and item occlusion?
- **RQ4:** How can models be lightweighted for real-time edge processing?

KEY CONTRIBUTION

Establishes a highly accurate (98.16%) 12-class waste vision benchmark emphasizing dataset balancing, augmentation, and lightweight execution.

OVERVIEW

Lv et al. (2023) present a 3D spatial localization system for robotic grasping. Combining a RealSense D415 RGB-D depth camera, YOLACT instance segmentation, point-cloud extraction, and DBSCAN clustering, the system calculates oriented bounding boxes for precise 3D grasping coordinates at up to 150 FPS.

PROJECT OBJECTIVES

1. Perform real-time RGB instance segmentation using YOLACT.
2. Generate individual masks and extract corresponding point clouds using depth data.
3. Filter noise with DBSCAN clustering and compute oriented 3D bounding boxes for robotic grasping.

RESEARCH GAP

Most vision systems stop at 2D bounding boxes or image classification, failing to provide the 3D spatial pose and orientation required for real-world automated robotic grasping in complex environments.

RESEARCH QUESTIONS

- **RQ1:** How effectively can instance segmentation isolate garbage items in complex scenes?
- **RQ2:** How accurately can RGB-D data determine 3D spatial position and orientation?
- **RQ3:** How does DBSCAN clustering improve point-cloud localization accuracy?
- **RQ4:** How can spatial coordinates be translated into robot base frame coordinates for grasping?

KEY CONTRIBUTION

Pioneers high-speed (150 FPS) 3D spatial pose estimation for garbage items by coupling real-time instance segmentation with DBSCAN-filtered point-cloud processing.